



ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

TECHNICAL SCIENCES P1
MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 10 Technical Sciences Paper 1 practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Penalise once per question for missing or wrong units. Give follow-through marks for correct use of an earlier wrong answer.

QUESTION 1

1 1.1 B ✓✓ 1.2 B ✓✓ 1.3 C ✓✓ 1.4 B ✓✓ 1.5 C ✓✓ [10]

QUESTION 2

2.1 (a) kilogram (kg) ✓ (b) second (s) ✓ (c) ampere (A) ✓ (3)

2.2 (a) 2 500 N ✓ (b) 0,45 m ✓ (c) $72 \div 3,6 \checkmark = 20 \text{ m} \cdot \text{s}^{-1} \checkmark$ (4)

2.3 $3,2 \checkmark \times 10^{-4} \text{ m} \checkmark$ (2)

2.4 $28 \times 0,01 = 0,28 \text{ mm} \checkmark$; $5,5 + 0,28 \checkmark = 5,78 \text{ mm} \checkmark$ (3)

[12]

QUESTION 3

3.1.1 $12 + 5 = 17 \text{ km} \checkmark$ (1)

3.1.2 $12 - 5 = 7 \text{ km} \checkmark$ north ✓ (2)

3.1.3 $20 \text{ min} = \frac{1}{3} \text{ h} \checkmark$; $17 \div \frac{1}{3} \checkmark = 51 \text{ km} \cdot \text{h}^{-1} \checkmark$ (3)

3.1.4 $7 \div \frac{1}{3} \checkmark = 21 \text{ km} \cdot \text{h}^{-1} \checkmark$ north ✓ (3)

3.2.1 The rate of change of velocity ✓✓. (2)

3.2.2 $30 = 10 + a(5) \checkmark \checkmark \rightarrow a = 4 \text{ m} \cdot \text{s}^{-2} \checkmark$ (3)

3.2.3 $\Delta x = \frac{10 + 30}{2} \times 5 \checkmark \checkmark = 100 \text{ m} \checkmark$ (3)

3.3 A scalar has magnitude only, e.g. distance, speed, time ✓; a vector has magnitude and direction ✓, e.g. displacement, velocity, force ✓. (3)

[20]

QUESTION 4

4.1 $w = mg = 200 \times 9,8 \checkmark = 1\,960 \text{ N} \checkmark$ (2)

4.2 $W = F\Delta x = 1\,960 \times 3 \checkmark \checkmark = 5\,880 \text{ J} \checkmark$ (3)

4.3 Power is the rate at which work is done (energy is transferred) ✓✓. $P = \frac{5\,880}{12} \checkmark \checkmark$
 $= 490 \text{ W} \checkmark$ (5)

- 4.4 $E_p = mgh = 5\,880\text{ J}$ ✓✓ (2)
- 4.5 $P_{\text{in}} = \frac{490}{0,7} \text{ ✓✓} = 700\text{ W}$ ✓ (3)
- 4.6 Energy cannot be created or destroyed; it can only be changed from one form to another ✓✓. Motor: electrical energy → mechanical (kinetic) energy ✓. (3)

[18]

QUESTION 5

- 5.1 Longitudinal: particles vibrate parallel to the direction of the wave ✓, e.g. sound ✓. (4)
Transverse: particles vibrate perpendicular to the direction of the wave ✓, e.g. waves on a rope, water waves, light ✓.
- 5.2 $v = f\lambda = 200 \times 1,7 \text{ ✓✓} = 340\text{ m}\cdot\text{s}^{-1}$ ✓ (3)
- 5.3 Ultrasound pulses are sent into the metal; a crack reflects the waves back early ✓, and the time of the echo shows where the crack is ✓. (2)
- 5.4 Very loud sound (about 85 dB and above) ✓ over long periods damages the hair cells in the inner ear ✓, causing permanent hearing loss ✓. (3)

[12]

QUESTION 6

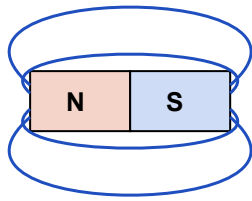
- 6.1 The opposition to the flow of electric current in a conductor ✓✓ (ratio of potential difference to current). (2)
- 6.2 $\frac{1}{R_p} = \frac{1}{6} + \frac{1}{12} \text{ ✓} = \frac{3}{12} \text{ ✓} \rightarrow R_p = 4\ \Omega$ ✓ (3)
- 6.3 $2 + 4 = 6\ \Omega$ ✓ (1)
- 6.4 $I = \frac{V}{R} = \frac{12}{6} \text{ ✓✓} = 2\text{ A}$ ✓ (3)
- 6.5 $V = IR = 2 \times 2 \text{ ✓} = 4\text{ V}$ ✓ (2)
- 6.6 $V_{\text{parallel}} = 12 - 4 = 8\text{ V}$ ✓; $I = \frac{8}{12} \text{ ✓} = 0,67\text{ A}$ ✓ (3)
- 6.7 $P = I^2R = 2^2 \times 2 \text{ ✓✓} = 8\text{ W}$ ✓ (3)
- 6.8 A fuse is a thin wire connected in series with the appliance ✓. If the current becomes too high, the wire melts ✓ and breaks the circuit, preventing damage, overheating or fire ✓. (3)

[20]

QUESTION 7

7.1

(3)



Marks: closed field lines from N around to S ✓; lines do not cross and are closest together at the poles ✓; arrows from N to S outside the magnet ✓

7.2

Any TWO: increase the current ✓; increase the number of turns of the coil ✓; use a soft-iron core

(2)

7.3

Any ONE: lifting scrap metal (crane) ✓ / relays and switches / electric bells / motors / separating iron from other waste

(1)

7.4

Any TWO: iron ✓; nickel ✓; cobalt; steel

(2)

[8]

TOTAL: 100

END OF MARKING GUIDELINE